

Replication Plan

Physics-Based Hybrid Machine Learning for Critical Heat Flux Prediction with
Uncertainty Quantification

OSTI ID: 2571909

Auto-generated Replication Blueprint

April 8, 2026

Contents

1	Paper Overview	2
2	Detailed Workflow	2
2.1	Phase 1: Data Acquisition and Preprocessing	2
2.2	Phase 2: Baseline Physics Correlations	2
2.3	Phase 3: Hybrid ML Model Building	2
2.4	Phase 4: Uncertainty Quantification	3
2.5	Phase 5: Validation	3
3	Datasets	3
4	Models, Tools, and Software	3
4.1	Hardware Requirements	4

Paper Overview

This paper develops a physics-based hybrid machine learning approach for predicting critical heat flux (CHF) in nuclear reactor thermal-hydraulics. The hybrid model combines established empirical CHF correlations (e.g., Groeneveld look-up table, W-3 correlation) as physics-based features with data-driven models (deep neural networks, Bayesian neural networks, deep Gaussian processes) to improve prediction accuracy. Uncertainty quantification (UQ) is provided through Bayesian approaches. The model is trained and validated on the NRC/EPRI CHF database.

Detailed Workflow

Phase 1: Data Acquisition and Preprocessing

1.1 Obtain the CHF database.

- NRC/EPRI CHF database or the publicly available subset.
- Features: pressure, mass flux, quality, heated length, hydraulic diameter, etc.
- Target: critical heat flux (CHF) value.

1.2 Filter and clean data.

- Remove outliers and entries with missing values.
- Apply parameter range restrictions as in the paper.

1.3 Feature engineering.

- Compute physics-based features: correlation predictions (W-3, look-up table, etc.).
- Normalize/standardize inputs.

1.4 Split data into train/validation/test sets (ratios as specified).

Phase 2: Baseline Physics Correlations

2.1 Implement CHF correlations.

- Groeneveld 2006 look-up table.
- W-3 correlation.
- Other correlations referenced in the paper.

2.2 Evaluate correlation performance on the test set (RMSE, R^2 , MAPE).

Phase 3: Hybrid ML Model Building

3.1 Deep Neural Network (DNN).

- Input: physical features + correlation predictions.
- Architecture: fully connected, 3–5 hidden layers, ReLU activation.
- Train with Adam optimizer, MSE loss.

3.2 Bayesian Neural Network (BNN).

- Variational inference or MC-Dropout for uncertainty estimation.
- Output: predictive mean and variance.

3.3 Deep Gaussian Process (DGP).

- Stacked GP layers for capturing non-stationary relationships.
- Inference via variational methods (e.g., doubly stochastic VI).

Phase 4: Uncertainty Quantification

4.1 Compute predictive intervals (e.g., 95% credible intervals).

4.2 Calibration analysis: compare predicted vs. observed coverage.

4.3 Decompose uncertainty: aleatoric vs. epistemic components.

Phase 5: Validation

5.1 Compare hybrid vs. pure-ML vs. pure-correlation approaches.

5.2 Reproduce error metrics (RMSE, MAE, R^2) from paper tables.

5.3 Reproduce parity plots, residual distributions, UQ coverage plots.

Datasets

Dataset / Source	Type	Location / Notes
NRC/EPRI CHF database	Experimental	Contact NRC or EPRI for access; some subsets available in literature
Groeneveld 2006 look-up table	Correlation table	Published in: Groeneveld et al., Nuclear Engineering and Design (2007)
Paper's data splits	Specified	Training/test splits described in paper methodology

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
PyTorch	≥ 1.10	DNN and BNN implementation
GPyTorch	≥ 1.9	Deep Gaussian process https://gpytorch.ai/
Pyro	≥ 1.8	Probabilistic programming for BNN https://pyro.ai/
scikit-learn	≥ 1.0	Data preprocessing, metrics, baseline models
NumPy / SciPy	≥ 1.21	Numerical operations
Pandas	≥ 1.3	Data manipulation
Matplotlib / Seaborn	≥ 3.4	Parity plots, residual plots, UQ visualization

Hardware Requirements

- GPU recommended for BNN and DGP training (≥ 8 GB VRAM).
- ≥ 16 GB RAM.